

## Topics

- o Square roots in  $\mathbb{R}$
- o Decimals
- o Finite and countable sets

## Square roots in $\mathbb{R}$

Thm For any real  $y > 0$ , there exists a unique  $x > 0$  s.t.  $x^2 = y$ .

proof:

Uniqueness: suppose  $x_1^2 = y$  and  $x_2^2 = y$ .

$$x_1 < x_2 \Rightarrow x_1^2 < x_2^2, \text{ contradiction.}$$

$$x_2 > x_1 \Rightarrow x_2^2 > x_1^2, \text{ contradiction.}$$

We must have  $x_1 = x_2$ .

Existence: Let  $A = \{t \in \mathbb{R}_{>0} \mid t^2 < y\}$ .  $A$  is

- non-empty: For  $y \geq 1$ ,  $\frac{1}{2} \in A$ .

For  $y < 1$ , we have  $y^2 < y$ , so  $y \in A$ .

- bounded above:  $(y+1)^2 = y^2 + 2y + 1 > y$  so  $(y+1)^2 > t^2 \forall t \in A$ .

Thus,  $y+1$  is upper bound of  $A$ .

Let  $x = \sup A$ .

Suppose  $x^2 < y$ . Then  $(x + \varepsilon)^2 < y$  for  $\varepsilon = \frac{1}{2} \min\left(1, \frac{y - x^2}{2x+1}\right)$

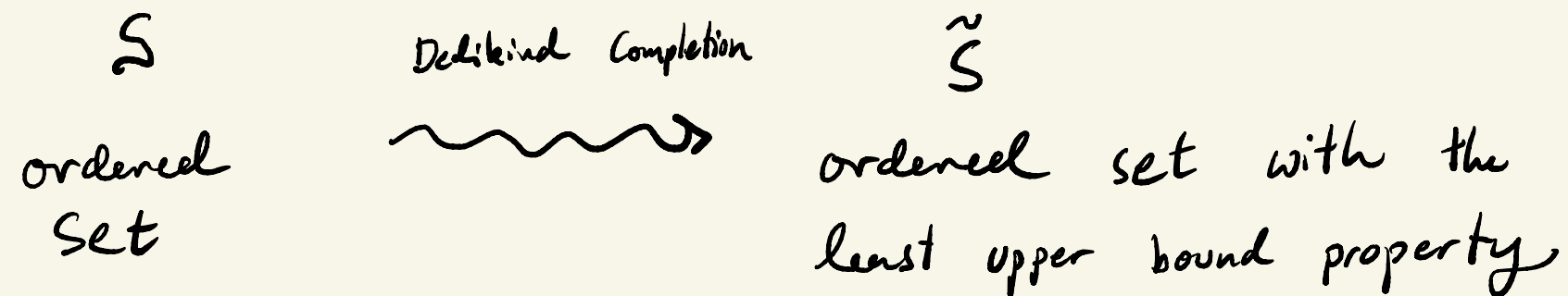
So  $x$  is not an upper bound of  $A$ .

Suppose  $x^2 > y$ . Then  $(x - \varepsilon)^2 > y$  for  $\varepsilon = \frac{1}{2} \min\left(x, \frac{x^2 - y}{2x}\right)$

So  $x$  is not the least upper bound.

This leaves only the possibility  $x^2 = y$ .

# Dedekind Completion (For culture)



Definition for  $S = \mathbb{Q}$

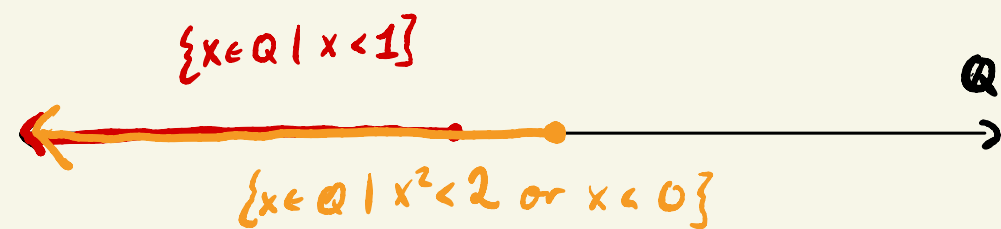
Def A cut is a subset  $\alpha \subset \mathbb{Q}$  s.t.,

I)  $\alpha$  is not empty

II) If  $p \in \alpha$ ,  $q \in \mathbb{Q}$ , and  $q < p$  then  $q \in \alpha$

III) If  $p \in \alpha$  then  $p < r$  for some  $r \in \alpha$ .

Def As an ordered set,  $\mathbb{R}$  is the set of cuts of  $\mathbb{Q}$ , with  $\alpha < \beta$  if  $\alpha \subsetneq \beta$ .



Thm (Rudin Ch1 Appendix)

- The Dedekind completion of  $\mathbb{Q}$  has the least upper bound property
- We can extend  $+$  and  $\cdot$  from  $\mathbb{Q}$  to  $\mathbb{R}$ , making it an ordered field.

How can we write down or store a real number?

## Decimals

Let  $x$  be real.

Let  $n_0$  be the largest integer s.t.,  $n_0 \leq X$ .

Let  $n_1$  be the largest integer s.t.,  $n_0 + \frac{n_1}{10} \leq X$

Let  $n_k$  be the largest integer s.t.,  $n_0 + \frac{n_1}{10} + \dots + \frac{n_k}{10^k} \leq X$

$n_0 . n_1 n_2 n_3 \dots$  is the decimal expansion of  $x$ .

$n_0 \in \mathbb{Z}$  ,  $n_i \in \{0, \dots, 9\}$  for  $i \geq 1$ .

Prop Decimal expansion is a bijection from  $\mathbb{R}$  to the  
Set of sequences  $n_0, n_1, n_2, \dots$  with  $n_0 \in \mathbb{Z}$  and  $n_k \in \{1, \dots, 9\}$   
Such that the sequence isn't eventually all 9s.

Proof Idea Given such a sequence  $n_0, n_1, \dots$  we get a real number

$$X_{(n_0, n_1, \dots)} = \sup_{n \in \mathbb{N}} \left( \sum_{k=0}^n \frac{n_k}{10^k} \right). \quad \text{We should show this is inverse to decimal expansion.}$$

Storing a real number requires infinite memory.

Def Let  $f: A \rightarrow B$  be a map of sets.

- The image of  $E \subset A$  is

$$f(E) = \{f(a) \mid a \in E\}.$$

- The preimage of  $E \subset B$  is

$$f^{-1}(E) = \{a \mid f(a) \in E\}.$$

We write  $f^{-1}(b)$  for  $f^{-1}(\{b\})$  for  $b \in B$ .

Def Let  $A$  and  $B$  be sets

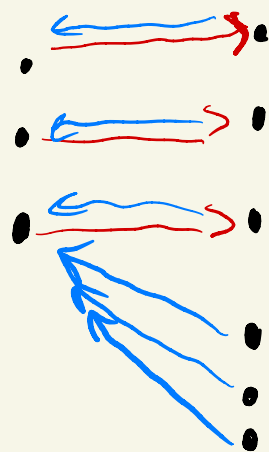
- $A \sim B$  if there exists a bijection  $A \xrightarrow{\sim} B$ .  $A, B$  have same cardinality.
- $A \leq B$  if there exists an injection  $A \hookrightarrow B$ .  $A$  has smaller cardinality than  $B$ .

Lemma For any two sets  $A, B$

$\exists$  injection  $A \hookrightarrow B \Leftrightarrow \exists$  surjection  $B \twoheadrightarrow A$

Proof sketch

From an injection,  
build a surjection the other way



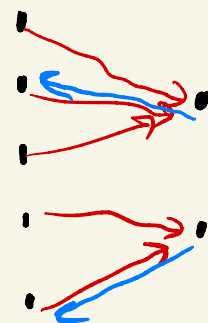
Let  $f: A \hookrightarrow B$  be injective

Choose arbitrary  $a_0 \in A$

Let  $g(b) = \begin{cases} a \text{ s.t. } g(a) = b & \text{if } b \in f(A) \\ a_0 & \text{if } b \notin f(A) \end{cases}$

$g$  is surjective.

From a surjection  
build an injection the other way



Let  $f: A \twoheadrightarrow B$  be surjective

Choose  $a_b \in f^{-1}(b)$  for each  $b \in B$

(uses axiom of choice)

Let  $g(b) = a_b$  for all  $b \in B$ .

$g$  is injective.

Theorem (Cantor-Schröder-Bernstein)

$$\bullet A \leq B \text{ and } B \leq A \Rightarrow A \sim B$$

Try to prove it! (not so easy)

This shows bijection classes of sets are an ordered set.

Def Let  $\underline{n} = \{1, \dots, n\}$ .

Def  $A$  is finite if  $A \sim \underline{n}$  for some  $n \in \mathbb{N}$ .

We say  $|A| = n$ , and  $A$  has cardinality  $n$ .

Example If  $|A| = m$ ,  $|B| = n$  then

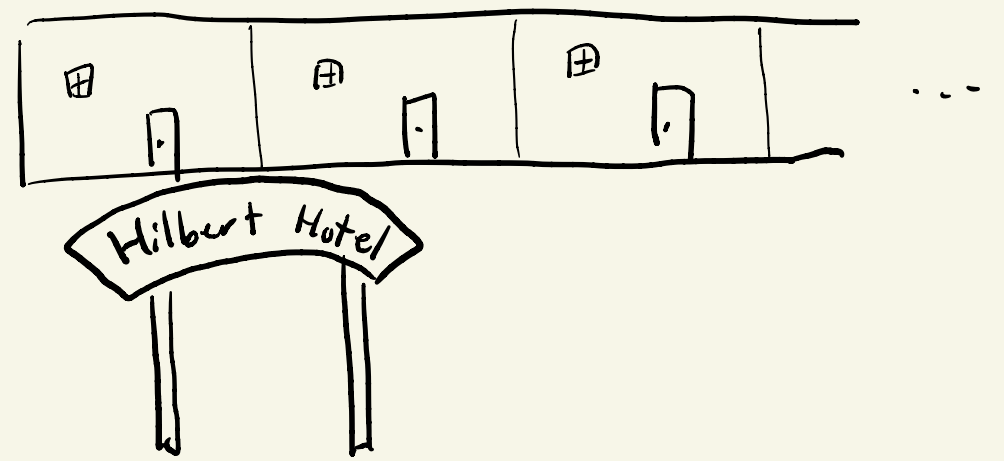
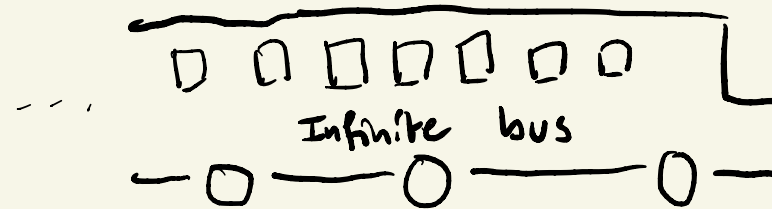
- $|A \times B| = mn$  product set
- $|A \cup B| \leq m + n$  union
- $|P(A)| = 2^m$  power set  $P(A) = \{\text{subsets of } A\}$

Example  $\mathbb{Z} \sim \mathbb{N}$

$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}$$

$$\phi: \mathbb{N} \rightarrow \mathbb{Z}$$

$$\phi(n) = \begin{cases} -\frac{n-1}{2} & n \text{ odd} \\ \frac{n}{2} & n \text{ even} \end{cases}$$



Def  $A$  is countable (countably infinite) iff  $A \sim \mathbb{N}$ .

Def A sequence of elements in a set  $A$  is a function  $\phi: \mathbb{N} \rightarrow A$ .

We write  $x_n = \phi(n)$ , and call the sequence  $x_1, x_2, x_3, \dots$  or  $(x_n)$ .

*Slight difference from Rudin's notation  $\{x_n\}$ .*

"Arrange  $A$  into a sequence" means choose a bijection  $\mathbb{N} \rightarrow A$ .